Diksha Jangam

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Summary

MSc Robotics student at the University of Birmingham who builds autonomous systems end to end, from sensing and perception through planning to control and testing. Implemented SLAM, path planning and trajectory tracking from scratch, integrated LiDAR and camera pipelines in ROS 2, and built an Arduino/ESP32 obstacle-avoidance vehicle in embedded C/C++. Tests systems against timing budgets and injected sensor noise, and explains technical results clearly to engineers and non-specialists alike.

Technical Skills

- **Autonomy & Navigation:** SLAM (log-odds occupancy grids, ORB-SLAM3, slam_toolbox), A*, Pure Pursuit, trajectory generation, odometry + IMU fusion, finite state machines
- **Perception:** YOLOv8, ByteTrack, OpenCV, ArUco visual servoing, depth back-projection, LiDAR and camera integration, Open3D, COLMAP
- **Simulation & Robotics Software:** ROS 2 Humble (nodes, launch, TF2, rosbag), Gazebo (camera, LiDAR, IMU plugins), Webots, NVIDIA Isaac Sim (setup), sensor-noise injection
- **Embedded & Hardware:** Arduino, ESP32, embedded C/C++, sensor-driven control logic, serial and gRPC/protobuf communication
- **Machine Learning:** PyTorch, TensorFlow, CNN training, DINOv2 feature probing
- **Tools:** Python (primary), C/C++, Bash, Linux/WSL, Git, Docker, GitHub Actions, pytest (50+ cases)

Education

M.Sc. Robotics, University of Birmingham Sept 2025 – Present

Motion Control & Path Planning, Intelligent Robotics, Industrial Automation

B.Sc. Data Science & Analytics, University of Mumbai Graduated 2025

CGPA 8.4/10, top 3 of 70 students; ML, Deep Learning, Computer Vision, Data Structures

Robotics & Autonomy Projects

Autonomous Hazardous Zone Detection Using SLAM Dec 2025

Python, Webots, camera, LiDAR

- Implemented a SLAM engine from scratch (log-odds occupancy grid of 50 m × 50 m at 0.1 m, Bresenham ray tracing, odometry + IMU gyroscope fusion) to correct heading drift on uneven terrain.
- Built a hysteresis-based state machine separating static obstacles from human workers using camera and LiDAR, and injected sensor noise to test robustness.

ROS 2 Path Following System Mar 2026

ROS 2 Humble, Webots, OpenCV, Streamlit

- Developed a three-node navigation stack (path smoothing, trapezoidal-velocity trajectories, Pure Pursuit tracking) running at 40+ Hz with under 30 ms total cycle time.
- Optimised the control loop from 15 to 40+ FPS (2.67×), added LiDAR obstacle avoidance and HSV worker detection (95% accuracy), and built a live Streamlit dashboard.

Object-Based SLAM for Warehouse Automation Apr – Aug 2026

MSc Thesis; ROS 2, Gazebo, YOLOv8, ByteTrack, ORB-SLAM3, LiDAR

- Planned the thesis as a 12-week phased programme with a formal risk assessment, then built a ten-package ROS 2 system combining a trained YOLOv8n detector, ByteTrack tracking and depth back-projection into a 3D object map.
- Validated every camera detection against an independent LiDAR ray-cast before acceptance; processed 3.69×10^5 tracking events and reduced 1.44×10^5 mapping records to a 28-landmark map.
- Traced four misleading metrics to root causes, including a camera-intrinsics mismatch and ground-truth odometry masking a real trajectory error.

Autonomous Obstacle-Avoidance Vehicle

Arduino, ESP32, embedded C/C++

- Built a physical obstacle-avoidance vehicle on Arduino and ESP32, programming real-time sensor-driven navigation logic in embedded C/C++.

Autonomous Grasp System Jun 2026

ROS 2 Humble, PyTorch, OpenCV, ikpy, Docker, GitHub Actions

- Built an end-to-end pipeline (ArUco visual servoing, CNN grasp-quality predictor, 6-DoF inverse kinematics) coordinated by a finite state machine.
- Benchmarked each stage against latency budgets (vision under 33 ms, inference under 100 ms, IK under 50 ms) and wrote 50+ pytest cases with GitHub Actions CI and Docker.

CNN Lane-Following Controller Oct 2026

PyTorch, ROS 2 Humble, Gazebo, OpenCV

- Built a CNN steering controller from camera images, running closed-loop predictions at 30 Hz in a Gazebo warehouse simulation with 3 human obstacles.
- Generated 500 synthetic training samples and retrained for 10 epochs (training loss 0.597 to 0.013).

LLM-Driven Navigation Agent May 2026

Python, Webots, A, LLM APIs*

- Built an agent in which an LLM proposes 4 candidate routes from natural-language instructions and a second LLM judges them, executed with A* on a 0.2 m inflated occupancy grid.
- Removed coordinate hallucination by snapping model outputs to valid goals and added exponential-backoff retries.

Physical Inconsistency Detector

DINOv2, PyTorch, multi-view geometry, synthetic data

- Built a DINOv2 residual-probe detector with multi-view and temporal-sync consistency checks on synthetic data, reaching a multi-view AUROC of 0.94 and raising downstream mean IoU from 0.405 to 0.459.

3D Scene Reconstruction from Video

COLMAP, Open3D, Python

- Reconstructed 3D scenes from video using COLMAP structure-from-motion and Open3D point-cloud processing, working with camera poses and registration.

Touchless Hand-Tracking Mouse

Python, OpenCV, MediaPipe

- Built a touchless mouse controlled by hand tracking from a live webcam feed using OpenCV and MediaPipe.

NVIDIA Isaac Lab and Isaac Sim Setup

Isaac Lab, Isaac Sim, Python

- Configured the Franka Panda robot's joint setup in NVIDIA Isaac Lab and built a warehouse simulation environment in Isaac Sim.

Experience

Data Science Intern, Erohsik Scientific Instruments Apr – Jun 2024

- Built a tested Python pipeline with schema validation and quality checks over 5000+ records, cutting processing time by 22%.

Data Infrastructure Intern, SDFine Chem Limited Jun – Jul 2023

- Designed a cloud data warehouse and ETL pipelines for 8 stakeholder use cases, cutting decision latency by 4 hours.

Certifications

RoboDK Training (robot simulation and programming, 7-course series) | AWS Builders Online Series 2023